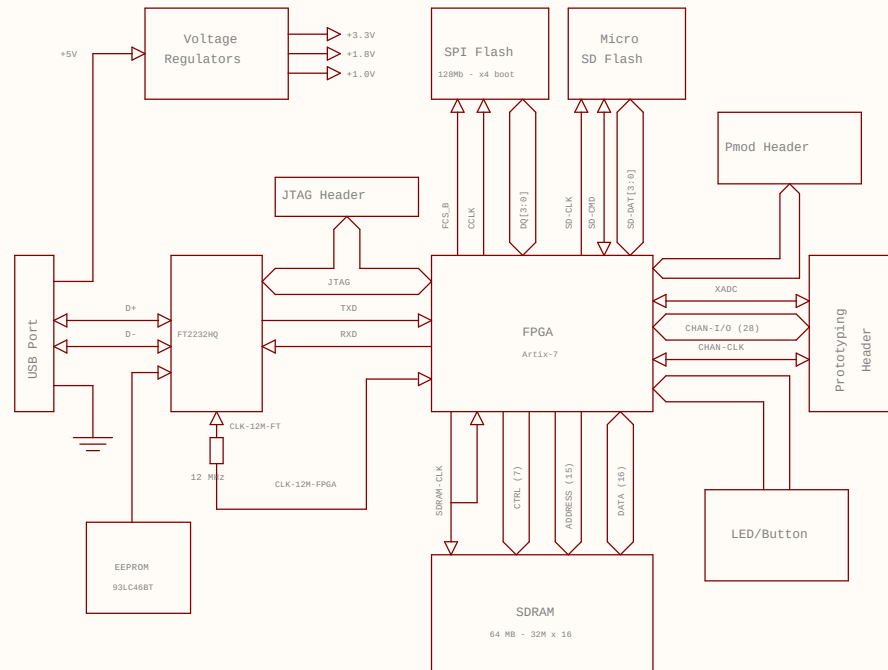
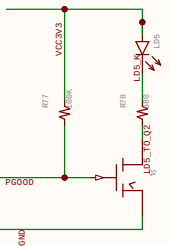


SPI flash sits on the FPGA's dedicated master-SPI config pins: CCLK C11, FCS_B K19, DQ[3:0] on D18/D19/G18/F18. Read at power-up.

microSD runs the native 4-bit SD bus, not SPI: CLK, CMD and DAT[3:0] on six bank-34 balls of its own, so a card cannot touch the config bus.

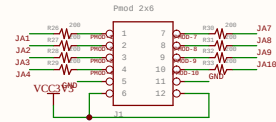
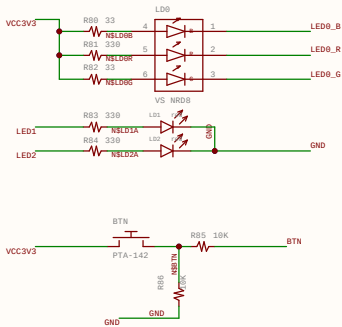
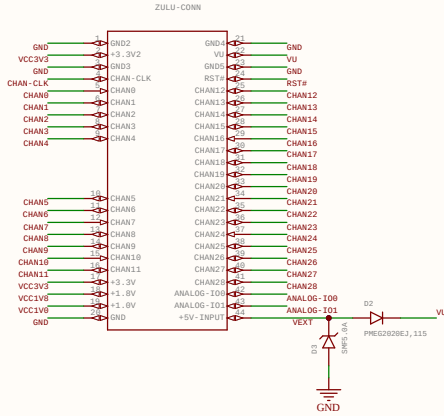


Block Diagram			
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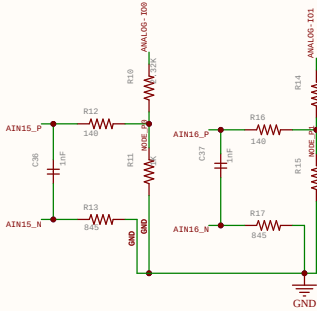


Power Supplies			
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X2: 44 pins on a 2.54 mm grid, rows 0.700 in apart, board 69.85 x 20.32 mm.
Numbered right to left and straight through: top row 1-20, bottom row 21-44.
TOP ROW, x 50.69 down to 1.27: GND, +3.3V, GND, then CHAN-CLK and CHAN0..CHAN3
(pins 4-8), CHAN4 (pin 9), CHAN5..CHAN11 (10-16), +3.3V, +1.8V, +1.0V, GND.
FOUR positions between pin 9 and pin 10 -- x 26.82, 34.29, 31.75, 29.21 -- are
RESERVED, not spare: the landing for the edge-mounted micro-USB X1, 7.80 mm of
copper in 11.176 mm clear, centred with 1.69 mm each side.
BOTTOM ROW, contiguous, x 50.69 down to 1.27: GND, VU, GND, RST#, CHAN12..CHAN28
(pins 25-41), ANALOG-I00, ANALOG-I01, +5V-INPUT.
FIVE grounds: pins 1, 9, 20, 21, 23. Pin 22-VU SOURCES 5 V, pin 44 only takes it.

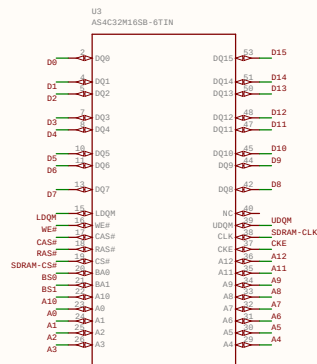
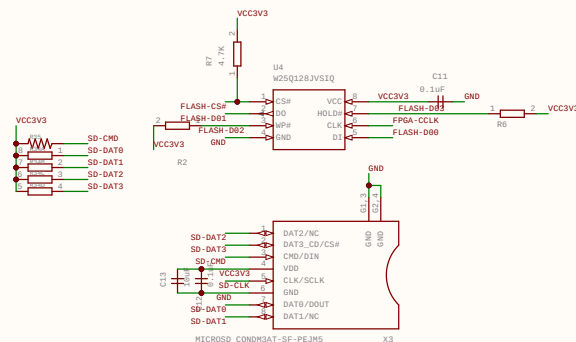


Digilent 12-pin Pmod (JA). Pins 1-4 and 7-10 are signals through 200R series resistors; pins 5/11 are GND and 6/12 are VCC3V3 -- so a 2x6 header carries EIGHT signals, not twelve, which is why the names skip JA5/JA6. No decoupling here: the pmod spec puts bypassing on the module, and VCC3V3 already carries ~170uF. The Cmod A7 has none at its Pmod either. JA1-JA4 (the top row, the half a Pmod uses for SPI) are all in bank 16: A14 A15 B15 A16. JA7 is C15, also bank 16. JA8/JA9/JA10 are W16/W2/W7 -- banks 14/35/34 -- because nothing else was free. All five banks are VCC0=3.3V, so this costs timing coherence, not levels.

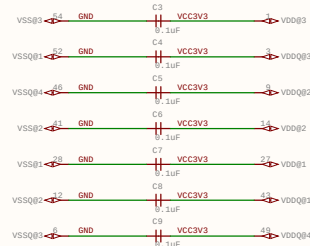


General IO			
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U3 = AS4C32M16SB-6T1N: 512Mbit (32M x 16) = 64MB.
Pin-for-pin and dimensionally identical to the 256MB
AS4C16M16A-6T1N it replaces -- all 54 pins and every
TSP. T1 dimension checked against both datasheets, and
pin 40 is NC on both.
ONE DIFFERENCE, AND IT IS NOT A PIN: the column address
is A0-A9 (1024 columns), not A0-A8 (512). The memory
controller must be set to 10 column bits, or it reaches
only half the array. Row A0-A12, banks BA0/BA1 and
refresh (8192 cycles / 64ms) are unchanged.
IDD1 rises 60mA -> 120mA, on the 600mA VCC3V3 buck
(LTC3569 SW3) -- worth a rail budget check.



microSD runs the NATIVE 4-BIT SD BUS, not SPI. CLK, CMD and DAT0-DAT3 -- six signals, all in bank 34 byte group 0 (R2 T1 T2 U1 V2 W2), so they can be constrained as one bus. DAT1/DAT2 came from prototyping-header CHAN1/CHAN0; DAT3 is the old SPI chip-select, which needs no new pin. R34/R35 are the 4.7k pull-ups the SD spec requires on CMD and DAT0-DAT3. R9 (100R on DAT0 R34) was removed: one damped line out of four is a skew, not termination.

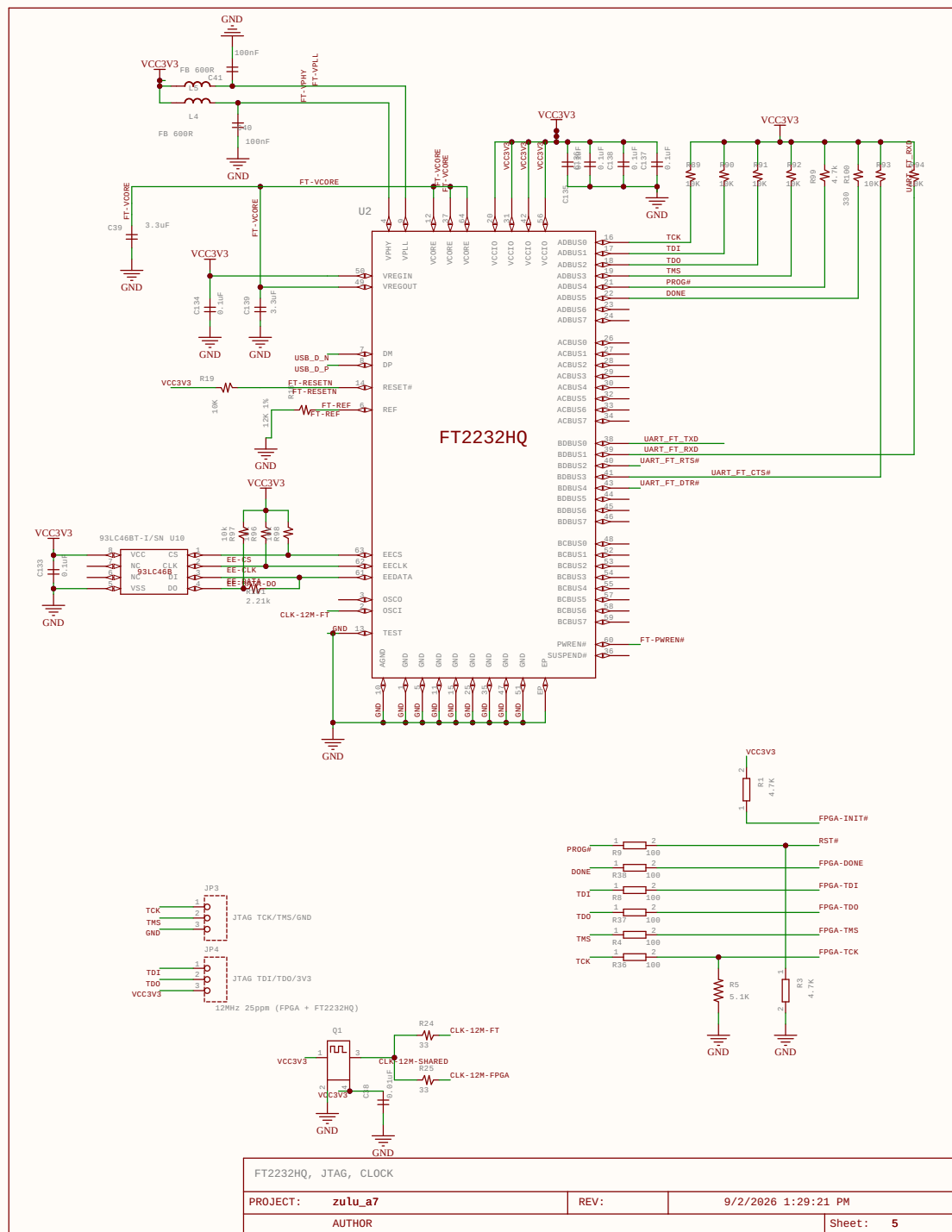


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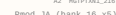
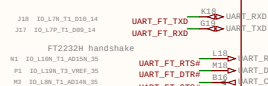
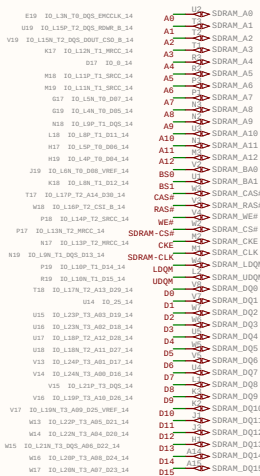
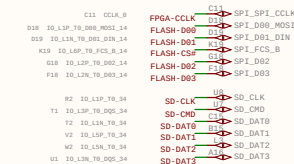
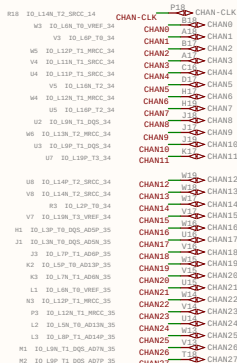
U3 DECOUPLING -- each cap bridges one supply pin to the ground pin OPPOSITE it,
and the other TSPIN-1 pin faces pin 55- and ever, supply pin faces the ground pin of its
own domain: VDD0 1/4/1/7 to VSS 5/4/3/2/1, VDDQ 3/6/5/4/3 to VSSQ 5/2/4/6/1/2, allance
isolates VDDQ from VDD for Q0 noise immunity. -- LAYOUT must KEEP THE PAIRS: each cap of
the underside spanning its two pins, own vss, VDDQ returning to VSSQ and VDD to VSS.
C10 is bulk and pairs with nothing. do NOT split the ground plane.

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Memory			
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PROOFREADING HEADER



FPGA Connections			
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